STEM Fellowship Big Data Challenge 2022

Investigating the Correlations Between International Wealth Inequalities and Renewable Energy

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Abstract

Access to renewable energy is becoming increasingly important in the face of climate change and pollution. However, this transition may be more difficult for disadvantaged groups such as those of a low socio-economic status. In this study, we compared the socio-economic status of various countries to their generation, production and investment in renewable energy in order to see if there was any sort of correlation. Our goal was to find a way to balance the importance of renewable energy with that of equality and standard of living for low-income families. Determinants of such socio-economic status included the Human Development Index, which measures quality of life, and the Gini Index, which measures income inequality. Python and R were used as tools to find significant statistical correlations as well as to plot graphs and regression lines. The overall low P-values and high R-values of these results indicate a significance in the correlations between the chosen responding variables and renewable energy access / production. Most often, we saw upward trends when comparing our explanatory variables to the Gini Index, indicating higher income inequality with increased focus on clean energy. The same increase of clean energy led to lower HDIs (lower overall quality of life). We found that there was often a higher correlation with HDI than with Gini Index, which shows that standard of living is more affected by changes in renewable energy than income inequality. This gives an idea of how the world can balance standard of living with renewable energy access. [1]

Keywords: Renewable energy, equality, socioeconomic status, Gini Index, Human Development Index

1 Introduction

Rising temperatures, melting ice caps, and the gradual degradation of the ozone layer are just a few realities that we face in the status quo—and for the most part, they stem from carbon-based energy sources such as natural gas and oil.[2] Due to the urgency of climate deterioration, it is increasingly critical for individuals to gain unrestricted access to clean energy to counteract the implications of climate change that will ultimately affect all of humankind. However, during this transition, a small portion of people with economic leverage will have unobstructed access to clean energy, while the rest of the population will lose fundamental aspects of their economies, as they depend more on industries relating to non-renewable energy. [3]

We want to expand on this disparity at an international level, and analyze how access to clean energy can affect or correlate to inequality in various regions of the world. Our end objective is to do three things: raise more awareness about such issues affecting the lower-class, predict standard of living and inequalities in relation to specific renewable energy targets, and balance lowered standard of living with better environmental actions. We initially hypothesized that our three main areas of study—consumption, generation and investment—would all correlatee to an increase in Gini index and decrease in HDI.

2 Materials & Methods

2.1 Data Collection

By using public datasets from sources such as Our World in Data, we were able to conduct a

thorough analysis.

The Renewable Energy Database, an Our World in Data compilation by Hannah Ritchie and Max Roser [4] provided us with multiple datasets on renewable energy consumption and generation. [5] [6] [7] [8] To find the percentage of installed energy capacity that is generated, we compared the energy generation of a country against its capacity. We gathered our energy generation information from a dataset in the database mentioned above, titled "Renewable energy generation" [9]. It includes overall renewable energy generation for each country.

As for energy consumption, we used a database titled "Share of final energy consumption from renewable sources" [10], also compiled by Hannah Ritchie and Max Roser. This describes the percentage of overall energy consumption of renewable sources such as hydropower, solar energy, etc.

Two other databases were used as quality of life calculators: the Gini Index [11] and the Human Development Index [12].

The Gini Index or **Gini coefficient** is used to find and compare income inequalities between countries. It was used to find a correlation between energy access and wealth disparity. In our graphs, each Index was multiplied by 100 to get a value between 1 and 100. A higher Gini index is a higher level of income inequality.

The Human Development Index (HDI) is a statistical index that measures a country's development through factors such as a healthy lifestyle, access to education, Gross National Income per capita, and life expectancy. It is measured as a number between 0 and 1, with a higher HDI correlating to a higher quality of life.

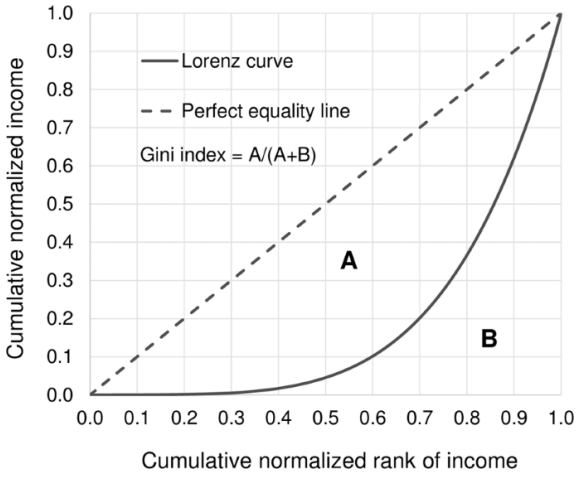


Figure 1: This is a graph of a Lorenz curve, where A is the area between the Line of Equality and the country's Lorenz curve, and B is the area below the Lorenz curve.

2.2 Tools and Resources

We were able to conduct a thorough analysis using the programming languages of R v4.1.0 (using RStudio), and Python v3.9 (using the application PyCharm).

The libraries in R that we used includes tidyverse, corrplot, Hmisc, psych, moonBook, ggiraph, ggiraphExtra, plyr, and magrittr.

The following libraries in Python we used include Matplotlib, matplotlib.pyplot, and pandas.

We also used Excel, Google Sheets, and the Power Tools add-on in order to clean some of our raw data in preparation for analysis.

2.3 Data Analysis

The first layer of basic dataset selection (filtering and graphical representation) included initial data cleaning to remove unusable data points and filtering datasets to meet our needs.

In the second layer, we analyzed different sources of data and compared their correlation and relevancy with each other. From there, we constructed a correlation comparison to see which pairs were the most strongly tied and visualized them again.

Next was the problem of how we chose our statistical information. We wanted statistics to describe change (slope), strength of correlation (R value), and statistical significance (P-value). **Pearson's Correlation Coefficient (R)** is used to represent the strength and direction of the association between two variables.

Last, we debated between LASSO regression, ridge regression, logistic regression, and least squares regression as our graphical model. We chose least squares as it seemed to fit our data the best and was the most accessible. **Least squares regression lines (lines of best fit)** use predictive features to construct and visualize the linear correlation between two variables. **Multiple regression lines** are quite similar, but they have the capacity to compare 2 or more predictor values.

3 Results

3.1 Prerequisites

First, to make sure there was no multicollinearity (a correlation between two independent variables), we built a plot between our two main independent variables, renewable energy consumption and generation [Figure 2]. Fortunately, the R value of only 0.0382 indicates almost no multicollinearity. This adds validity to

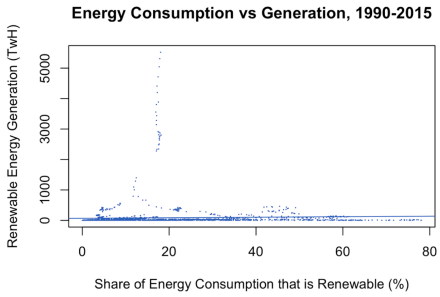


Figure 2: This graph was made to disprove strong multicollinearity between renewable energy consumption and generation. Renewable energy consumption was placed on the X axis (measured using a percentage compared to total energy consumption) while renewable energy generation was placed on the Y axis (measured in terawatt-hours (TWH)). The R value was 0.03821731, representing a very weak positive relationship between the two variables.

our results, showing that there will not be any overlap.

3.2 Energy Usage

After the initial steps, we began analyzing the data that we had retrieved. We started with the analysis of energy consumption, comparing a country's energy consumption to Gini Index and Human Development Index.

In Figure 3, the least squares regression line generated a slope of 0.10154—for every increased percentage point of renewable energy consumption, the corresponding Gini Index value increased by that much. The R value of 0.3418 shows a moderate correlation between the two [see section 4.3] and the low P-value of the line of best fit gives the results statistical significance. Similarly, the residual standard error is only 7.5, meaning that, on average, actual datapoints only differ from the predicted value (line of best fit) by a Gini Index of 7.5. Next, the F statistic of 16.01 creates a ratio of the variances between the two most extreme degrees of freedom. The farther away from 1 the F statistic is, the more significant the variables in the graph are. Our large F statistic conveys the significance of both variables. From these statistics, we can conclude that an increased focus on renewable energy correlates to an increase of the Gini index, which aligns with our hypothesis.

Next, by plotting the relative amount of energy consumption against each country's HDI [Figure 4], we can compare a country's renewable energy consumption to its standard of living. The least squares regression line follows a

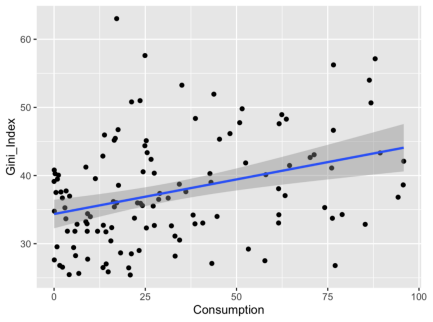


Figure 3: In this scatter plot, various countries' consumption of renewable energy (measured using percentages) were plotted against their Gini Indices to form a least squares regression line (blue line). Each data point on the graph represents one country (although some are missing), and all the data came from 2015, the most recent year found in both datasets. The shaded grey around the blue line of best fit represents the confidence interval for the line.

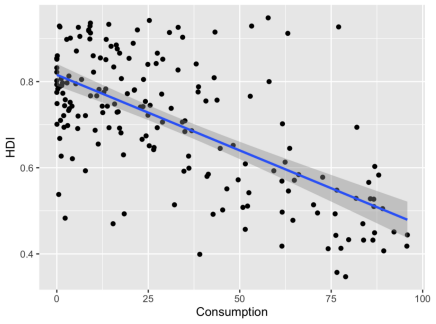


Figure 4: This scatter plot compares a country's percentage of energy consumption that is renewable (X axis) to its Human Development Index (Y axis). Each point on the graph is the data found in 2015 for one country, 2015 being used because it was the most recent year found in both datasets. The least squares regression line is shown in blue, and its confidence interval is represented by the shaded gray area.

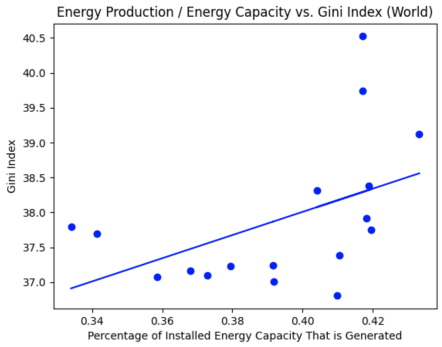


Figure 5: A scatter plot that compares energy generation-capacity (X axis, measured in percentage points) to Gini Index (Y axis), where each point on the graph represents the global average for a single year. The years between 2000 - 2017 (although not all) were plotted on this graph as they were the most recent points found in both datasets. The blue line represents the corresponding least squares regression line.

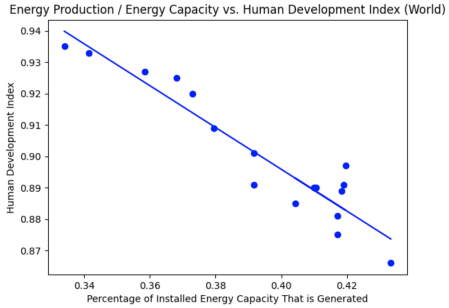


Figure 6: This graph uses a scatter plot to represent the relationship between energy generation-capacity (%) and the Human Development Index. Each point on the graph is the global average for a year, ranging from 2000 - 2017. A least squares regression line was used to model the data.

slope of -0.0035, meaning the HDI decreases (signifying a decrease in living conditions) by that amount for every percentage point of renewable energy consumption that increases. This slope is significant, given the small range of possible HDI values. The R value of -0.64605 indicates a strong, negative correlation between the two variables. The P-value of this graph is nearly 0; the results are of great significance. The F statistic of 133.2 on 1 and 186 df indicates a very strong significance of both variables on the graph—even stronger than in figure 3. The relationship is reliable, but the standard residual error (0.1175) is high; the HDI could vary over a large range. Overall, however, this graph indicates that greater clean energy consumption is directly associated to a worse HDI, once again strengthening our hypothesis.

3.3 Energy Generation

Next, we compared the world's energy capacity to standard of living indicators for various consecutive years to see if there was any relation between the two.

The first step was to compare the standard of living indicators with the percentage of the respective country's capacity to generate renewable energy. We did this by comparing two datasets (capacity and production) and dividing the former by the latter. The capacity dataset lacked a lot of datapoints, resulting in a very small dataframe [see section 4.3].

Figure 5 shows the relationship between energy generation and Gini Index. The R-value (0.4743) indicates a moderate positive relationship between the two variables, while the slope of the line (16.58) indicates a large increase in Gini index (increased inequality) for a small increase in energy generation. The low P-value of the graph (0.0543) suggests that the results are significant. Through these statistics, we found that increased renewable energy generation correlates to greater income inequality, further supporting our hypothesis.

Next, we compared the same explanatory variable (production / capacity) to HDI. Again, we found that very few datapoints were common between all 3 datasets, so there weren't as many points as in other graphs.

The regression line for Figure 6 represents the negative correlation between energy generation and the Human Development Index (both global averages). This is supported by the R-value of -0.9438, indicating a very strong negative correlation between the two. The P-value is extremely small, so these results are statistically significant. This graph indicates that globally, a lower HDI is the result of an increase in renewable energy generation.

These results align with our initial prediction, that more focus on renewable energy would correlate to decreased standard of living.

3.4 Energy Investment

Finally, we analyzed energy investment. Similar issues from a lack of common data points were faced for both of the graphs generated.

Figure 7 analyzes the relationship between energy investment and the Gini Index for 3 major countries. The slope of -2.2069e-10 shows almost no change in the Gini Index with a change in renewable energy investment, but this is logical since a one dollar increase in investment would

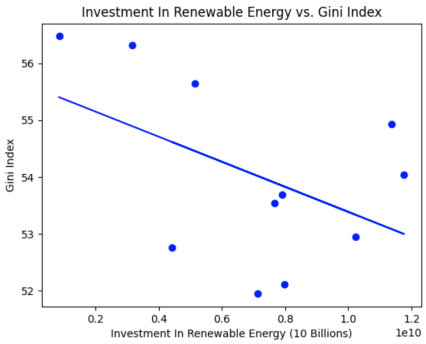


Figure 7: This graph compares investment in renewable energy in $ 10 billions USD (X axis) to the global average of Gini coefficients from 2004 - 2015. Each data point represents a country: the United States, Brazil, or China. A least squares regression line is shown in blue.

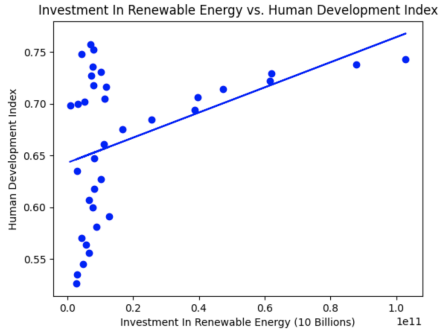


Figure 8: This scatter plot was created to compare investment in renewable energy (X axis, $10 billions USD) to the Human Development Indices coefficients for 3 countries (the United States, Brazil, and China). The line of best fit is shown in blue.

have little impact. A larger scale, like millions of dollars, would produce a larger slope. The R-value of -0.4700 suggests a moderate negative correlation, so the variables are related on a significant enough scale for the results to be considered valuable. The P-value is somewhat high, reflecting a high probability of these results occurring by chance. All the statistics indicate that increases in renewable energy investment correlates to decreases in Gini Indices and therefore, inequality.

Figure 8 compares renewable energy investment and HDIs in America, Brazil, and China. The slope shows an extremely small change in HDI with an increase in renewable energy investment. This has the same explanation as Figure 7. The R-value of -0.4199 indicates a moderate correlation between the two variables, and the P-value of this graph was extremely small, confirming the validity of the results. This relationship in this graph helped us conclude that renewable energy investment increases when standards of living are higher.

4 Discussion

4.1 Discussion between Responding Variables

The Gini Index and the Human Development Index were used as responding variables in our graphs. The Gini Index measures income inequality and the HDI measures standard of living, so we expected the slopes to move in opposite directions (with more renewable energy, the HDI would decrease while the Gini index would increase). For the most part, the results of the graph corresponded with our hypotheses from the introduction.

In almost all of our results, the graphs that included the Gini Index had positive slopes, indicating that greater income inequality was a result of increased renewable energy generation and consumption. The graphs comparing energy consumption and generation to HDI had negative slopes, suggesting that a lower standard of living can be explained by high energy generation and consumption.

When comparing the R values for these variables, a greater correlation is shown for graphs involving HDI. In 3.2, the graph with HDI (Figure 4) had a higher R value than the graph with Gini Index as its responding variable [Figure 3]. Coupled with its low P-value, we think that the results of Figure 4 may have more significance than that of Figure 3, meaning that HDI was affected more greatly than Gini index. The R value for Figure 5 suggests a moderate positive relationship between energy generation-capacity and Gini Index, but the R-value for Figure 6 (using HDI as its responding variable) is much stronger at nearly -1. Once again, the P-value for the graph with HDI is much smaller than the P-value for the graph with the Gini Index, indicating greater statistical significance for the results of Figure 6. When the variables are compared to energy investment, there is a break in pattern. While the slope of graphs with Gini Index and HDI were positive and negative respectively, this is switched in the energy investment analysis. However, this makes sense given that we measured the explanatory variable in dollars, instead of a proportion—countries with more money to invest will generally have better conditions and less inequality. Furthermore, the difference between the two R values show that the investment vs Gini Index graph [Figure 7] has a slightly stronger correlation between the

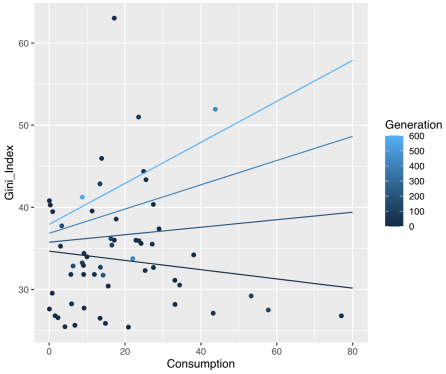


Figure 9: A scatter plot with multiple linear regression that compares percent of energy consumption that is renewable and energy generation in terawatt hours to the Gini Index. The scatter plot includes predictions of Gini Indices at different combinations of energy generation and consumption. We see a general upward trend in Gini index.

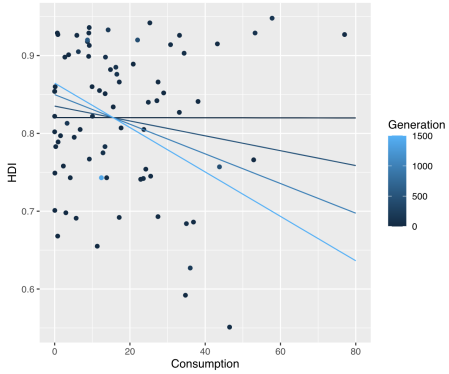


Figure 10: A scatter plot with multiple linear regression that compares percent of energy consumption that is renewable and energy generation in terawatt hours to the Human Development Index. The scatter plot includes predictions of HDI at different combinations of energy generation and consumption. We see a general downward trend in HDI.

two variables than the investment vs HDI graph [Figure 8]. Still, the P-values of the graphs follow the aforementioned pattern: a higher significance for Figure 8, the graph involving HDI.

Overall, from the comparisons of statistics, the graphs involving HDI always had a higher significance than the graphs with Gini Indices, and the R-values indicated stronger correlations. The HDI graphs tended to have positive slopes, while the Gini Index graphs had negative ones, with the exception of the graphs in 3.4. With these results, it appears that HDI may be more significantly affected by differences in energy generation, capacity, and investment. This may be because the Human Development Index includes a variety of standards, including GDP, life expectancy, and knowledge measured by years of schooling. The Gini Index has a much smaller scope, as it only focuses on income inequality.

4.2 Overall Discussion

Figure 11 is a correlation matrix that presents the R-value for each relationship, as well as density plots and histograms for each dataset. However, these results should only be used to compare different relations against each other and never considered by itself, as the small amount of datapoints most likely led to inaccuracies and unusually high values.

The top half of squares show that, generally, the strongest relationship is between energy generation and HDI, while the weakest relationship is between energy consumption and Gini index. This, once again, supports the earlier conclusion that HDI is affected more greatly than Gini index.

4.3 Sources of Errors

Almost all of our outputs had an R value that was either relatively weak or medium-strength. This is largely due to two facts: 1) It is almost impossible to find a high R value when comparing two real-world datasets due to the presence of lurking variables that weaken correlations. 2) We had relatively small datasets since the initial layer of data cleaning removed a lot of datapoints not common to all sets.

Having small datasets could also be the source of error for other statistical factors, such as high residual standard errors, etc.

The last possible source of error could be the type of regression we chose for specific models. In Figure 5, we could've tried out an exponential model, and in Figure 8, a logistics growth model could've better fit the datapoints, resulting in a more accurate prediction. We chose not to fit those specific models due to inconvenience and time restraints.

Conclusion

As seen in the results and discussion, the most significant results came from HDI comparisons to percentage of installed energy capacity that is generated. It seems that more renewable energy access correlates with higher income inequality and a lower standard of living, as we predicted in our hypothesis. Future studies could use the results of this project to further discover how to

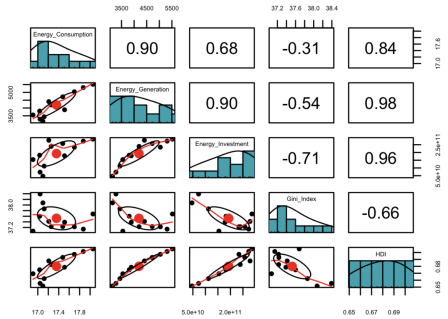


Figure 11: A correlation matrix between each of our datasets, with R values at the top, histograms in the middle, and density plots at the bottom. This plot compares the R and P-values of the relationships. Many relationships have fairly low P-values, highlighting the accuracy of the results. The high R-values support the strength of the correlations between the chosen variables.

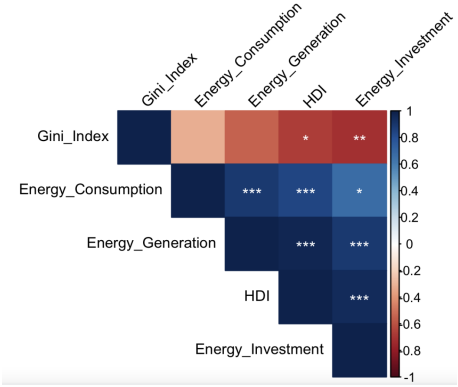


Figure 12: A correlation matrix heat graph showing R-values (colour) and P-values (astericks). * means a P-value of less than 0.05, ** means less than 0.01, and *** means less than 0.001. Blank squares means that the P-value does not fall under any of these categories.

achieve the ideal energy consumption and generation without detracting from quality of life—a way to balance and optimize the two. With this objective in mind, one can build off of this study and go into greater detail of how to increase energy access for lower-class people or marginalized groups. The results of this project, along with potential future studies, will help determine specific solutions for how to solve a major global crisis.

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